

Tracking Spatiotemporal Mineral Dissolution Through Quantitative Nuclear Magnetic Resonance Signatures

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Key Points:

- Nuclear magnetic resonance tracks spatiotemporal carbonate dissolution dynamics in pore-scale reactive transport simulations
- Distinct dissolution regimes produce diagnostic transverse relaxation pathways over time
- Earlier pore coupling induced by dissolution enables more efficient permeability breakthrough

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Abstract

Mineral dissolution reorganizes pore networks through spatiotemporally variable pathways, but non-invasive tracking remains difficult. We test whether nuclear magnetic resonance (NMR) signatures can diagnose carbonate dissolution by integrating pore-scale reactive transport simulations with forward modeling of T_2 distributions for face dissolution, wormholing, and channeling regimes. Simulated T_2 pathways show dissolution progress is controlled by fluid exchange between newly formed vugs and the matrix. Face dissolution evolves from single peak to separated peaks and back to single peak as coupling weakens and the matrix is consumed. Wormholing maintains a broad single peak as competing pathways merge, whereas channeling shows transient bimodality followed by rapid peak merging as a dominant conduit expands. A matrix-vug connectivity index derived from T_2 spectra, together with flow-field tortuosity, shows that earlier matrix-vug exchange coincides with faster tortuosity reduction and earlier permeability breakthrough. These results support quantitative NMR signatures as indicators of dissolution dynamics.

Plain Language Summary

When reactive fluids flow through rocks, they dissolve minerals and reshape the pore space. These changes do not occur everywhere in the same way. Dissolution may advance as a front, form a single efficient channel, or create several competing pathways before flow becomes strongly connected. Tracking these changes is important for applications such as carbon dioxide storage, groundwater protection, and subsurface energy extraction, but direct imaging is difficult during ongoing reactions. In this study, we use numerical simulations to test whether nuclear magnetic resonance (NMR), a non-invasive method sensitive to water in pores, can quantitatively track how dissolution evolves over time and space. We show that dissolution patterns leave distinct signatures in NMR signals. These signatures reveal whether dissolution is localized near the inlet, concentrated in one dominant channel, or distributed through competing flow paths. By analyzing these signals, we can estimate how well the original rock matrix pores connect to newly formed dissolution cavities. Our findings may expand the use of non-invasive NMR tools for monitoring the spatiotemporal evolution of subsurface pore structures and may improve predictions of flow dynamics in geological carbon sequestration and energy extraction.

Keywords

carbonate dissolution; nuclear magnetic resonance; transverse relaxation time; reactive transport; pore-scale simulation; permeability breakthrough

1 Introduction

Mineral dissolution reshapes pore networks through spatiotemporally variable pathways, influencing flow and transport evolution in many natural and engineering processes, including carbonate weathering and karst development (Kump et al., 2000; White et al., 1995), pyrite weathering in acid mine drainage (Nordstrom, 2009), biological carbonate dissolution induced by ocean acidification (Andersson & Gledhill, 2013), and geological carbon storage (Elkhoury et al., 2013; García-Ríos et al., 2015). Depending on the interplay of advection, diffusion, reaction, and formation structure, mineral dissolution may proceed as uniform dissolution, face dissolution, wormholing, or channeling (Golfier et al., 2002; Lucas et al., 2023; Szymczak & Ladd, 2009). These regimes control dissolution morphology and how efficiently connected flow paths develop and permeability changes. When reaction is much slower than transport, the reactants penetrate the entire domain, leading to uniform dissolution (Roded et al., 2020). When transport is significantly slower than reaction, dissolution localizes near the inlet as a progressing face dissolution front (C. E. Cohen et al., 2008). At higher injection rates, reactive-infiltration instabilities focus flow and form wormholes (Fredd & Fogler, 1998; Z. Wang et al., 2022), whereas channeling rapidly expands preferential flow paths along their trajectory (Menke

et al., 2016). Predicting dissolution regimes under various conditions is important. For example, acidizing relies on the wormholing regime to maximize hydraulic conductivity while minimizing acid consumption. Conversely, applications such as carbon sequestration require preventing conductive preferential pathways, because unexpected channeling or wormholing can compromise storage integrity and induce leakage (A. Cohen et al., 2021; H. Deng & Peters, 2019; Elkhoury et al., 2013; García-Rios et al., 2015; Ott & Oedai, 2015; Steefel et al., 2013).

Extensive experimental and numerical studies have characterized dissolution induced pore structure changes, permeability evolution, and regime transition boundaries using acidizing experiments, Darcy- and pore-scale simulations, and time-lapse X-ray microtomography (μ CT) (Agrawal et al., 2021; Fredd & Fogler, 1998; Golfier et al., 2002; Hoefner & Fogler, 1988; Menke et al., 2015, 2016, 2018; Ott & Oedai, 2015; Y. Yang et al., 2020; J. Yang et al., 2026). These studies show how transport-reaction competition and initial pore heterogeneity produce uniform dissolution, face dissolution, wormholing, and channeling (Menke et al., 2023; Szawello et al., 2024). However, regime identification still commonly relies on visual or image-derived descriptions of evolving pore geometry, or on simulated flow fields and streamlines (Menke et al., 2018; Szawello et al., 2024). μ CT imaging resolves dissolution structures in detail, but it is typically limited to small extracted samples, requires offline scanning and image processing workflows, and is not suited for continuous, *in-situ* monitoring during ongoing subsurface reactions. Recently proposed quantitative approaches reduce reliance on visual classification but often require detailed flow-distribution information that is rarely accessible in experiments or field settings (Szawello et al., 2024). Geophysical methods can monitor mineral dissolution non-invasively over larger spatial and temporal scales by tracking changes in properties linked to porosity, saturation, pore-fluid conductivity, and elastic structure (Binley et al., 2015; Kuras et al., 2009; Berryman et al., 2002; Rembert et al., 2024). The central challenge is therefore to define process-based links between measurable geophysical signals and the spatiotemporal evolution of dissolution regimes.

Among these geophysical methods, low-field nuclear magnetic resonance (NMR) is particularly relevant as it directly responds to hydrogen-bearing pore fluids and can estimate water content, pore size distribution, fluid properties, and permeability (Behroozmand et al., 2015; Maillet et al., 2022; Xu et al., 2023). The transverse relaxation time (T_2) is widely used in hydrological investigations (Behroozmand et al., 2015; Knight et al., 2012; Sanderlin et al., 2013; Stingaciu et al., 2009), and T_2 monitoring has been applied to track pore structure evolution during carbonate dissolution (Mahmoud et al., 2016; Mahmoud, 2017; Seyyedi et al., 2020). During dissolution, newly formed vugs, residual matrix pores, and developing conduits modify fluid and magnetization exchange between pores (here referred to as pore coupling). This pore coupling can alter the shape, width, peak position, and peak separation of the T_2 distribution (Soto Bravo & Zhang, 2023; Zhou et al., 2026), complicating direct pore size interpretation while also carrying information on evolving matrix-vug connectivity. Existing interpretations, however, mainly describe changes in apparent pore size or porosity (Mahmoud et al., 2016; Mahmoud, 2017). This is insufficient for distinguishing dynamic dissolution regimes that may produce similar bulk porosity increases but develop through different spatial pathways and breakthrough mechanisms. A forward-modeling framework is therefore needed to link reactive transport evolution with NMR responses and to interpret changes in T_2 signals over time as quantitative indicators of dissolution regimes, connectivity, and permeability breakthrough.

To address this need, we develop an integrated pore-scale reactive transport and NMR simulation framework to examine whether changes in T_2 spectra over time can distinguish dissolution regimes and track spatiotemporal carbonate dissolution dynamics. Using face dissolution, channeling, and wormholing regimes generated by the reactive transport model, we simulate bulk T_2 distributions and spatially resolved T_2 profiles from evolving pore geometries. We then compare T_2 evolution with permeability and flow-field

tortuosity to assess NMR’s potential as a proxy for the development of flow pathways and transport parameters. Finally, we introduce the matrix–vug connectivity (MVC) index to quantify matrix–vug coupling and link transient T_2 responses to permeability breakthrough. Together, our results indicate the potential of NMR as a non–invasive tool for monitoring pore structure evolution, characterizing dissolution regime transitions, and inferring hydraulic properties during carbonate dissolution.

2 Methods

Figure 1 summarizes the integrated framework used to link dissolution dynamics with NMR responses. The reactive transport module resolves transient pore geometry, flow, and solute concentration fields during acid–driven carbonate dissolution (Figure 1a,b). At selected dissolution stages, the resulting pore geometries are passed to the NMR forward model to generate whole–domain bulk T_2 distributions and spatially resolved T_2 profiles (Figure 1c). This synchronization enables direct comparison among pore–structure evolution, hydraulic properties, and NMR observables.

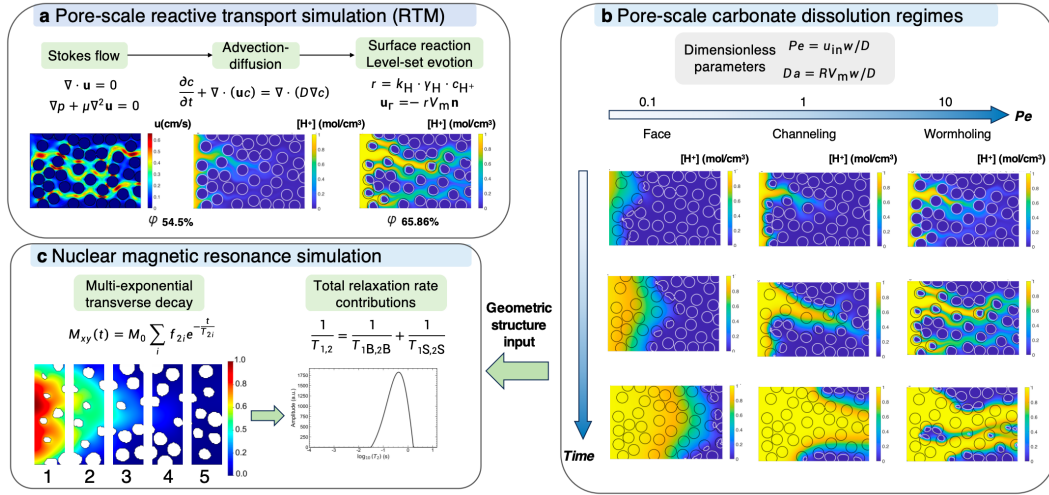


Figure 1: Coupled reactive transport and NMR framework for carbonate dissolution. (a) Governing equations for flow, solute transport, reaction, and interface retreat. (b) Dissolution regimes controlled by Péclet (Pe) and Damköhler (Da) numbers. (c) NMR simulation and T_2 inversion for whole–domain and five–segment analyses. In the concentration maps, white circles denote current CaCO_3 grains and black circles denote their initial positions.

2.1 Pore–Scale Reactive Transport Simulation

We use an open–source two–dimensional pore–scale reactive transport model (RTM) based on the RTSPHEM simulation toolkit (FAU Erlangen–Nürnberg) to simulate acid–driven calcium carbonate dissolution (Frank, 2014; Gärttner & Frank, 2021; Gärttner et al., 2020; Ray et al., 2019). The model couples steady Stokes flow, hydrogen–ion advection–diffusion, first–order surface reaction, and level–set tracking of the solid–liquid interface, preserving pore–scale resolution of interface motion while keeping the computation tractable (Baquer & Chen, 2022; Gärttner et al., 2020; Meakin & Tartakovsky, 2009; Molins et al., 2021; Ray et al., 2019). Since pore–scale flows in subsurface porous media typically occur at low Reynolds numbers, inertial effects are neglected and fluid motion follows steady Stokes

flow (Molins et al., 2012, 2014, 2021). Full governing equations and numerical procedures are provided in Text S1 in Supporting Information S1.

Calcite dissolution was modeled as a first-order surface reaction (J. Deng et al., 2025; Molins et al., 2021; Szawello et al., 2024; Szymczak & Ladd, 2009), with surface hydrogen ion molar dissolution rate as

$$R(c_{H^+}) = k_{TST} \gamma c_{H^+}, \quad (1)$$

where k_{TST} is an effective dissolution flux coefficient and γc_{H^+} is the dimensionless activity term. The corresponding normal retreat velocity of the solid-liquid interface is

$$v_n = V_m R(c_{H^+}), \quad (2)$$

where V_m is the molar volume of calcite. This formulation captures the transport-reaction competition while remaining consistent with pore network and simplified reactive transport studies (J. Deng et al., 2025; Szawello et al., 2024).

The operating conditions are governed by the inlet velocity U , molecular diffusion coefficient D , surface reaction rate constant k_{TST} , initial solute concentration c_0 , activity coefficient γ , and characteristic length L . The competition between transport and reaction is characterized by the Péclet number and Damköhler number:

$$Pe = \frac{UL}{D}. \quad (3)$$

$$Da = \frac{\gamma c_0 V_m k_{TST} L}{D}. \quad (4)$$

where L denotes the representative pore-throat spacing. Pe measures the balance between advection and molecular diffusion, whereas Da measures the balance between surface reaction and diffusive transport. In the parameter sweep, Pe was varied mainly through U , and Da through c_0 , with D and k_{TST} fixed. The effective calcite dissolution flux coefficient was set to $k_{TST} = 10^{-4} \text{ mol cm}^{-2} \text{ s}^{-1}$ (Chou et al., 1989; Batchelor-McAuley et al., 2022; Molins et al., 2021). Because γc_0 is dimensionless in this formulation, k_{TST} has the units of a surface molar flux. The explored parameter ranges and representative regimes are summarized in Text S8 and Figure S6.

The computational domain $\Omega = [0, L_x] \times [0, L_y]$ was populated with random, non-overlapping circular calcite grains of fixed radius. The inlet prescribed a uniform injection velocity and reactant concentration, the outlet used zero-stress and natural outflow conditions, and the lateral boundaries were treated as no-slip walls for flow and no-flux boundaries for solute transport. The domain size, grain geometry, boundary and initial conditions, physicochemical parameters, and time-stepping settings are summarized in Table S1; the random-packing procedure is described in Text S2. This simplified geometry preserves the first-order connectivity features needed for pore-scale dissolution analysis at reduced computational cost (Menke et al., 2015; Molins et al., 2021). At each sampling step, pore-space and solid masks were exported to the NMR module with geometric and hydraulic indices, including porosity, solid-liquid interfacial area, grain volume, equivalent permeability, injected pore volumes, and flow-field tortuosity (Text S3).

The numerical setup was benchmarked against the microfluidic calcite dissolution experiment of Rembert et al. (2024). After normalizing for different initial porosity, the mean absolute difference between the experimental and simulated porosity growth progress was small, and the total time to complete solid dissolution was comparable (4.26 h experimentally and 4.28 h numerically). These results indicate that the model captures the first-order geometric evolution, interface loss timescale, and dominant grain retreat pathway, supporting its use for the scenario simulations (Text S5).

2.2 NMR T_2 Simulation

Low-field NMR relaxation arises from the return of hydrogen nuclear magnetization toward equilibrium after radio-frequency excitation. In porous media saturated with water,

the transverse relaxation time (T_2) is governed by bulk relaxation (T_{2B}), surface relaxation (T_{2S}), and relaxation induced by diffusion (T_{2D}) (Bloch, 1946; Brownstein & Tarr, 1979). The relaxation time is given by

$$\frac{1}{T_2} = \frac{1}{T_{2B}} + \frac{1}{T_{2S}} + \frac{1}{T_{2D}}. \quad (5)$$

In geologic media without internal magnetic field gradients, surface relaxation at active solid–fluid interfaces dominates, while inflow and outflow effects and flow–induced diffusion relaxation are negligible under the low–velocity conditions considered here (Foley et al., 1996; Hall et al., 2016; Hemminga et al., 1977; Mohnke & Klitzsch, 2010). As a result, equation (5) can be written as:

$$\frac{1}{T_2} \approx \rho_s \frac{S}{V}, \quad (6)$$

where S is the water–contacting pore surface area, V is the fluid volume, and ρ_s is the surface relaxivity. This provides the physical basis for converting pore structure evolution induced by dissolution into simulated T_2 responses.

For each reactive transport snapshot, we export dissolution geometry and simulate the Carr–Purcell–Meiboom–Gill (CPMG) sequence using a 2D finite–element solver for the Bloch–Torrey equation (Torrey, 1956) to obtain magnetization distributions over time, with surface relaxation at active solid–fluid boundaries and zero–flux conditions at the inlet and outlet (Figure 1c; Text S4). The resulting T_2 decay is expressed as a multi–exponential relaxation (Bloch, 1946; Brownstein & Tarr, 1979):

$$M_{xy}(t) = M_0 \sum_i f_{2i} \exp\left(-\frac{t}{T_{2i}}\right), \quad (7)$$

where $M_{xy}(t)$ denotes the transverse magnetization, M_0 is the equilibrium magnetization proportional to the detectable protons in the sample, and f_{2i} is the signal fraction relaxing with time constant T_{2i} , with $\sum_i f_{2i} = 1$. The simulated decay curves are inverted using consistent T_2 inversion settings for all dissolution stages (Text S4), ensuring that spectral changes reflect evolving pore geometry rather than processing artefacts.

The Bloch–Torrey equation, implementation details (e.g., the finite element discretization and mesh regeneration procedure), and simulation settings (e.g., heterogeneous geometry handling, boundary condition assignment, and automation details) are provided in Text S4 and Table S2.

To integrate the reactive transport and NMR simulation modules, each time–stamped reactive transport snapshot was exported as updated pore geometry with Pe , Da , characteristic length, and structural metrics derived from flow. The extracted pore–space and solid masks were converted into the NMR mesh, where CPMG T_2 decay signals were simulated and inverted to yield the bulk T_2 distribution and spatial T_2 profile. These outputs were paired with porosity, permeability, tortuosity, and other reactive transport metrics at the same dissolution stages, allowing direct comparison between structural evolution, hydraulic response, and NMR T_2 signatures.

3 Results and Discussion

Our simulations span Péclet ($0.001 \leq Pe \leq 50$) and Damköhler ($0.001 \leq Da \leq 10$) numbers to generate representative dissolution regimes; the full parameter map is reported in Text S8 and Figure S6. To examine how NMR signatures distinguish dissolution pathways and link them to breakthrough efficiency, we focus on three representative cases at fixed $Da = 0.0369$: diffusion–dominated face dissolution at $Pe = 0.1$, where acid consumption

near the inlet produces an approximately planar retreating front; channeling at $Pe = 1$, where preferential invasion establishes an early main conduit; and wormholing at $Pe = 10$, where stronger advection promotes competing and merging high-conductivity pathways (Fredd & Fogler, 1998; Hoefner & Fogler, 1988; Menke et al., 2016; Szawello et al., 2024; Szymczak & Ladd, 2009). This Pe range is consistent with previous pore-scale dissolution studies (Soulaire et al., 2017; Z. Wang et al., 2022); the selected regimes are representative examples, not a complete regime-boundary analysis. We also performed control simulations with ordered geometry to assess the structural sensitivity of these NMR interpretations (Text S9 and Figure S7). Figure 2a–c shows representative $[H^+]$ concentrations at comparable porosities ($\phi \approx 0.73$) for each regime. White circles mark current calcite grain boundaries and black circles their initial positions; the systematic retreat and disappearance of grains illustrate the extent of dissolution. The time evolution of these concentration fields is shown in Movies S1–S3.

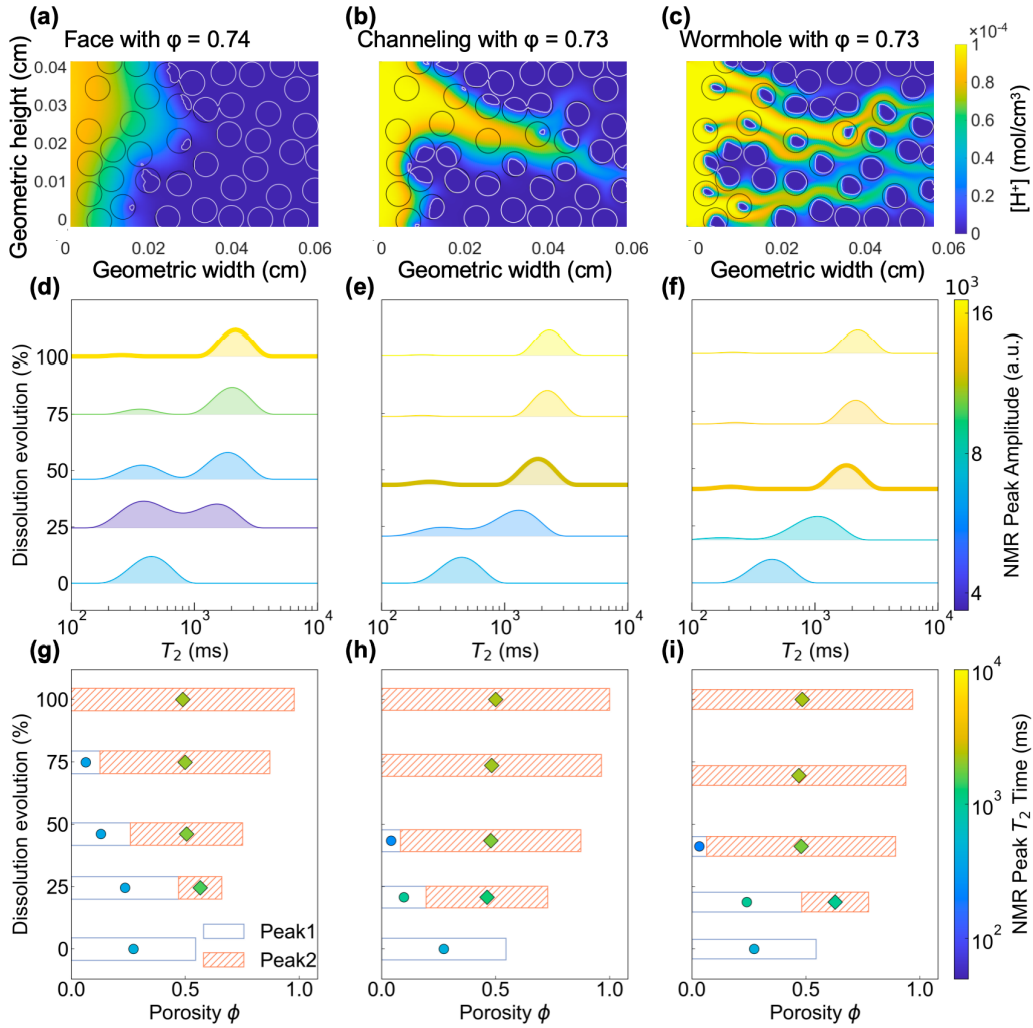


Figure 2: NMR inversion results for face dissolution, channeling, and wormholing. (a–c) Concentration snapshots at $\phi \approx 0.73$ for $Pe = 0.1, 1$, and 10 . (d–f) Inverted T_2 distributions at five dissolution stages; bold curves mark breakthrough time. (g–i) Matrix-pore and vug porosity partitions, with markers showing peak-center T_2 values. Columns correspond to $Pe = 0.1, 1$, and 10 from left to right.

3.1 NMR T_2 Response from Dissolution Patterns

The simulated T_2 spectra show that each dissolution regime follows a distinct relaxation pathway. Following previous carbonate NMR signal interpretation (Seyyedi et al., 2020), $T_2 \in [100, 1000]$ ms is treated as the matrix pore response, whereas values of $T_2 > 1000$ ms are associated with vugs formed by dissolution. Gaussian peak decomposition (Zhang & Zhang, 2021) is used to separate matrix and vug contributions where the spectra are partially overlapped (Figure 2g–i). Dissolution evolution is expressed as a percentage of the total dissolution duration, and the five displayed T_2 spectra correspond to 0%, 25%, 50%, 75%, and 100% of the normalized dissolution time for each case. The bold curves in Figure 2d–f mark spectra closest to permeability breakthrough, defined as the first time when $k/k_0 \geq 100$ (Kang et al., 2004).

In face dissolution ($Pe = 0.1$; Figure 2d), inlet-limited retreat separates newly formed vugs behind the reaction front from the largely unchanged matrix ahead (Figure 2a). This spatial separation converts the initial single matrix peak into a persistent bimodal T_2 spectrum: the short- T_2 matrix component shifts only modestly from about 450 to 357 ms, whereas the long- T_2 vug component shifts from about 1499 to 2171 ms as the dissolved inlet region expands (Figure 2g). Because the vug and matrix domains remain weakly coupled until the front approaches the outlet, breakthrough occurs late, at $\varphi = 0.98$ and $k/k_0 = 109.1$. The spectrum then returns to a single long- T_2 peak as the residual matrix signal disappears, consistent with face dissolution experiments reported by Mahmoud et al. (2016).

In contrast, the wormholing regime ($Pe = 10$) maintains a broad merged T_2 peak through most of dissolution (Figure 2f). Strong convection promotes concurrent growth, competition, and merging of multiple flow paths (Figure 2c), which sustains fluid exchange between matrix pores and vugs formed by dissolution. This coupling is reflected by the close decomposed matrix and vug peak centers at the second displayed stage, about 988 and 1035 ms, respectively (Figure 2i). Breakthrough occurs at $\varphi = 0.89$ and $k/k_0 = 106.0$, when the component associated with vugs already dominates the spectrum (92.7%).

The channeling regime ($Pe = 1$; Figure 2e) shows a shorter transition between these two responses. A principal channel penetrates early, focusing flow and producing transient T_2 bimodality as the local matrix and main conduit remain partially distinct. As this conduit widens along much of its length, exchange with adjacent matrix pores strengthens and the spectrum rapidly shifts toward the long relaxation component (Figure 2h). The vug contribution increases from 73.2% at 25% dissolution to 90.5% at 50%, immediately before breakthrough ($\varphi = 0.87$, $k/k_0 \approx 99.7$); the threshold is reached shortly afterward at $\varphi = 0.88$ and $k/k_0 = 107.7$. This transient bimodality followed by rapid peak merging indicates that channeling achieves early permeability breakthrough through efficient conduit formation with limited matrix dissolution.

Together, these results show that NMR T_2 spectra can distinguish the three dissolution regimes through their distinct pathways of flow reorganization. Face dissolution preserves a relatively stable matrix peak while progressively transferring signal to the vug peak. Channeling rapidly suppresses the matrix component as a single conduit expands. Wormholing keeps the two responses overlapped until the vug contribution dominates. Previous NMR studies on carbonate dissolution described changes in T_2 spectral morphology, showed that long T_2 components increase as large pores, vugs, or wormholes develop, and demonstrated NMR's capacity to assess wormhole matrix interconnectivity beyond CT or pressure drop measurements (Mahmoud et al., 2016; Mahmoud, 2017; Seyyedi et al., 2020). However, those studies did not define T_2 pathways for individual regimes over time. Our framework addresses this gap by clarifying the mechanisms of peak separation, overlap, and merging of NMR signatures as dissolution evolves, rather than treating the spectra only as apparent pore-size indicators.

We also calculated T_2 profiles, which provide spatially resolved information for applications such as stratigraphic delineation, lithological heterogeneity identification, and groundwater–table mapping (Griffiths et al., 2024; Vang et al., 2023). Here, we use these profiles (Text S6) to infer spatial geometric variations from NMR responses at permeability breakthrough across dissolution regimes.

3.2 Tortuosity Evolution Across Dissolution Regimes

Flow-field tortuosity τ is defined based on velocity ratios (Text S3) and provides an independent hydraulic reference to evaluate whether spectral evolution derived from T_2 reflects simplification of flow paths. Here, lower tortuosity indicates straighter and better connected flow pathways (Cai et al., 2019). Across all three dissolution regimes, global tortuosity generally decreases with increasing porosity (Figure 3c), from $\tau = 1.219$ at $\varphi = 0.55$ toward $\tau \approx 1$ near complete dissolution. However, the timing and pathway of this decrease differ among regimes, allowing direct comparison between T_2 evolution and simplification of flow paths.

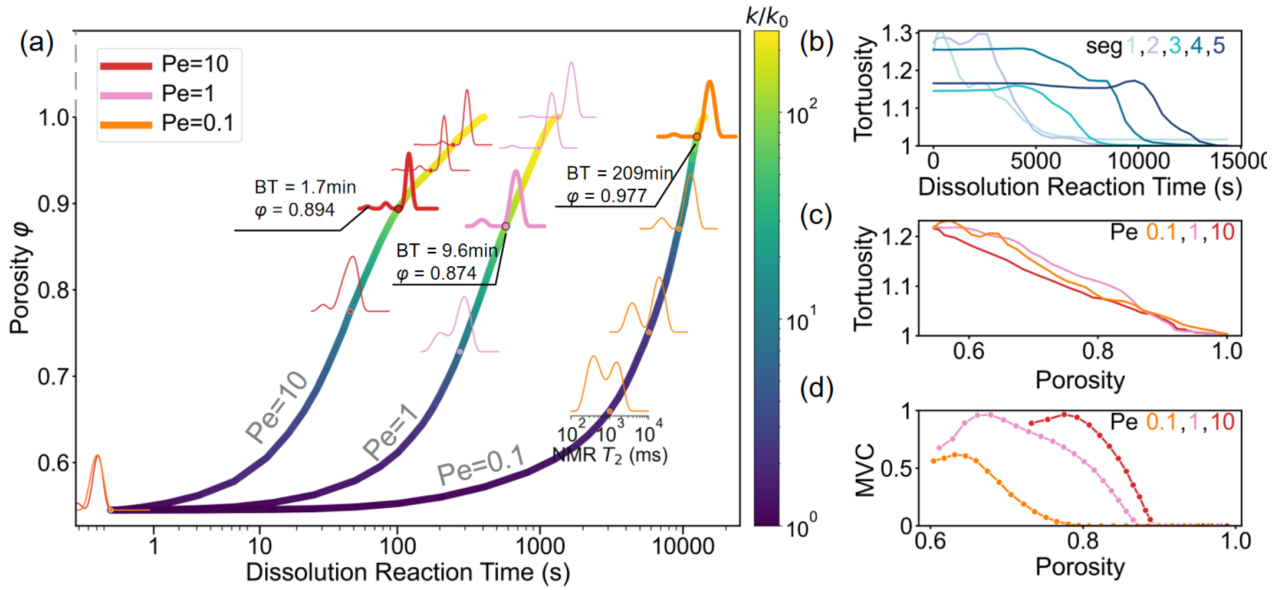


Figure 3: Porosity, permeability, tortuosity, and connectivity derived from NMR during dissolution. Amber, purple, and crimson denote $Pe = 0.1, 1$, and 10 in (a), (b), and (d). **(a)** Porosity trajectories colored by permeability ratio k/k_0 ; inset T_2 curves mark representative stages, with bold curves indicating breakthrough. **(b)** Segment-wise tortuosity for $Pe = 0.1$ from inlet to outlet. **(c)** Flow-field tortuosity as a function of porosity. **(d)** Matrix-vug connectivity metric (MVC) as a function of porosity.

Tortuosity provides a hydraulic check on the interpretation of flow path reorganization based on T_2 . In face dissolution, profiles by segment show a stepwise decrease in τ from inlet to outlet (Text S7; Figure 3b), indicating that the advancing vug region remains separated from residual matrix pores rather than forming an early coupled network. This delayed hydraulic connection is consistent with persistent T_2 peak separation and late breakthrough at $\varphi = 0.98$ ($\tau = 1.011$, $k/k_0 = 109.1$). In channeling, the sharp decline of τ to 1.103 by $\varphi \approx 0.82$ marks the stage after a principal conduit has penetrated the domain and begins to widen; the associated rapid permeability growth produces earlier breakthrough at $\varphi = 0.88$ ($\tau = 1.043$, $k/k_0 = 107.7$). This transition corroborates the transient T_2 bimodality followed

by peak merging, which records increasing exchange between the matrix and the expanding connected channel. In wormholing, τ is already similar at $\varphi \approx 0.88$ ($\tau = 1.045$), but k/k_0 remains 66.3 because competing branches have not yet coalesced into an equally effective connected pathway and global matrix–vug pore coupling develops later than in channeling. Breakthrough therefore occurs later at $\varphi = 0.89$ ($\tau = 1.038$, $k/k_0 = 106.0$). The influence of pore coupling on permeability breakthrough is quantified below using a metric derived from NMR discussed in the following section.

This observation is consistent with studies showing that dissolution reorganizes pore connectivity and tortuosity, thereby controlling permeability evolution (Luquot et al., 2014). Our results extend this relationship to NMR responses specific to each regime: delayed tortuosity reduction under face dissolution corresponds to persistent T_2 peak separation, whereas earlier tortuosity decline under channeling coincides with rapid peak merging. Previous NMR studies also showed that channel development induced by dissolution modifies diffusive transport and pore coupling (Elsayed et al., 2023; Muljadi et al., 2018; Oliveira et al., 2021). Compared with these approaches, our framework suggests that transient T_2 evolution can provide indirect evidence for hydraulic pathway reorganization during dissolution.

3.3 Quantification of Pore Coupling and Breakthrough Mechanisms

We next integrate the spectral, hydraulic, and trajectory information to test whether the regime distinction based on T_2 also explains the breakthrough hierarchy. Figure 3a embeds synchronized T_2 spectra along the porosity time trajectories and colors these trajectories by normalized permeability (k/k_0). This shows that the spectra are not isolated snapshots, but transient NMR responses tied to the same dissolution pathway that controls permeability growth. All cases begin from an early plateau near $\varphi \approx 0.55$, where k/k_0 remains close to 1, indicating that initial dissolution has not yet reorganized the pore network sufficiently to enhance flow. Because Pe was increased mainly by raising inlet velocity, the higher Pe cases receive stronger advective reactant supply relative to diffusion, which accelerates dissolution and permeability growth across the domain (Fredd & Fogler, 1998; Golfier et al., 2002; Szymczak & Ladd, 2009). The trajectories therefore brighten at very different rates: wormholing reaches the $k/k_0 \geq 100$ threshold (Kang et al., 2004) at about 1.7 min ($\varphi \approx 0.89$), channeling crosses the threshold shortly after the displayed stage at 9.6 min ($\varphi \approx 0.87$; first crossing at $\varphi \approx 0.88$), whereas face dissolution requires about 209 min and near total dissolution ($\varphi \approx 0.98$).

The colored trajectories further show that similar porosity does not imply a similar hydraulic state. Near $\varphi \approx 0.88$, face dissolution remains far from breakthrough ($k/k_0 \approx 8.9$), wormholing is approaching breakthrough ($k/k_0 \approx 66.3$), and channeling has already crossed the threshold ($k/k_0 \approx 107.7$). The embedded spectra explain this contrast: all regimes eventually shift toward long T_2 components as large pores and conduits grow, but their transient spectral pathways differ. Face dissolution preserves separated matrix and vug signals until late dissolution, wormholing maintains a broad merged response while multiple branches compete and coalesce, and channeling moves rapidly from transient bimodality to peak merging as one dominant conduit enlarges. This pathway dependence indicates that permeability breakthrough is governed less by final porosity than by when matrix pores and vugs formed by dissolution become hydraulically connected and diffusively coupled in the NMR response.

This trajectory view links the T_2 regime distinction to macroscopic breakthrough, but it does not directly quantify matrix–vug coupling. To extract this coupling information from the T_2 signal, inspired by Mahmoud et al. (2016), we define the matrix–vug connectivity metric (MVC) as

$$\text{MVC} = \frac{\min_{i \in [i_0, i_1]} S_i}{\max(S_{i_m}, S_{i_v})} \quad (8)$$

Here, S_i denotes the discrete T_2 amplitude at the i -th point; i_m and i_v are the dominant matrix ($T_2 \in [100, 1000]$ ms) and vug ($T_2 > 1000$ ms) peaks, respectively; and i_0 and i_1 define the valley-search interval between them. For weakly separated spectra, peaks from distinct pore systems are extracted using the Gaussian peak decomposition described above. MVC is evaluated only when matrix and vug components can be resolved or decomposed; fully merged spectra are interpreted together with peak migration and permeability evolution.

Physically, MVC tracks the degree of magnetization exchange between matrix pores and vugs formed by dissolution. Low MVC values denote strong peak separation and weak diffusive interaction, whereas increasing MVC indicates enhanced matrix-vug coupling expressed by shallower valleys or nearly merged T_2 peaks.

MVC resolves the pore coupling sequence behind the breakthrough hierarchy: channeling couples matrix and vug domains earlier than wormholing, whereas face dissolution maintains weak coupling until late dissolution. For channeling, the MVC maximum occurs at a relatively low porosity ($\varphi \approx 0.66$; Figure 3d). This early maximum reflects the formation of a single primary conduit. As this channel expands along its length, enhanced diffusive exchange develops between the conduit and adjacent matrix, consistent with early breakthrough. Conversely, the wormholing MVC peak ($\varphi \approx 0.77$) is delayed, suggesting that competition and coalescence among multiple pathways postpone the formation of a globally coupled network and defer permeability breakthrough. For face dissolution, the maximum MVC value remains low (≈ 0.63), subsequently diminishing to 0 as porosity approaches $\varphi \approx 0.79$. Driven by the progressing front and the pronounced disparity in pore-size distributions, the signal associated with the matrix decays before the component associated with vugs, reducing apparent exchange between the two populations. This weak coupling is consistent with the need for near-complete dissolution before hydraulic connection is achieved. Together with the earlier tortuosity decline in channeling, this MVC sequence suggests that early breakthrough is associated with both matrix-vug coupling and simplification of flow paths; wormholing is delayed because competing branches coalesce before forming an equally efficient hydraulic pathway. By tracking both T_2 pathways for each regime and MVC evolution, the proposed framework can distinguish dissolution modes and further separate channeling from wormholing by coupling sequence and breakthrough timing.

4 Implications and Significance

Our results suggest that NMR relaxation can provide more than a bulk porosity or permeability proxy during mineral dissolution. The evolution of T_2 spectra first distinguishes the three dissolution regimes through their contrasting peak separation, broadening, and merging pathways. Building on these spectral signatures for each regime, the combined analysis of matrix-vug coupling and breakthrough efficiency further distinguishes channeling from wormholing, which can have similar final conduit geometries but different coupling sequences and permeability trajectories.

This capability addresses a practical limitation in monitoring reactive porous media. Conventional NMR applications in subsurface engineering have primarily focused on bulk porosity and permeability estimation (Kenyon et al., 1988; Timur, 1969), whereas visual and image based pattern classification often emphasizes the final dissolution geometry. Recent MRI work on CO_2 injection into carbonate rocks has shown that magnetic resonance can map transport properties across different dissolution regimes (Fraboni, 2023); here, changes in T_2 responses over time, combined with the MVC index, provide a complementary way to track evolving matrix-vug coupling and distinguish channeling from wormholing by coupling sequence rather than by final geometry alone.

These findings may improve interpretation of NMR measurements in carbonate matrix acidification, CO₂-rich brine dissolution, and other reactive transport settings where pore connectivity evolves rapidly (Elkhoury et al., 2013; García-Ríos et al., 2015). They also provide a basis for integrating NMR with complementary geophysical methods such as spectral induced polarization (SIP) and self-potential (SP), which are sensitive to connected transport pathways and reactive interfaces (e.g., Cherubini et al., 2019; Placencia-Gómez et al., 2015; Rembert et al., 2023, 2025). Because the present framework is based on two-dimensional simulations with simplified reaction kinetics, use at the field scale will require three-dimensional validation, experimental calibration, and uncertainty assessment (Behroozmand et al., 2015; Jiang et al., 2020). More broadly, the same interpretation based on coupling may be useful for other reactive transport processes that reorganize pore topology, including mineral precipitation, biomineralization, and particulate clogging (Borer et al., 2026; Bottero et al., 2013; Carrel et al., 2018; J. Deng et al., 2025; Rege & Fogler, 1989; Yoon et al., 2012).

5 Conclusions

To our knowledge, this is the first framework that links pore geometry evolution during dissolution with NMR T_2 responses over time through coupled pore-scale reactive transport and NMR forward simulation. The results show that T_2 spectral evolution distinguishes the three dissolution regimes, while matrix-vug coupling and breakthrough efficiency further resolve differences between channeling and wormholing.

The three dissolution regimes produce distinct NMR relaxation evolutions. Face dissolution evolves from a single peak to separated peaks and then back to a single peak because the advancing vug region remains weakly coupled to the residual matrix. Channeling rapidly shifts from transient bimodality to peak merging as a single conduit penetrates and enlarges coherently, producing early breakthrough. Wormholing maintains a broad merged response but requires competing branches to coalesce before efficient breakthrough. The MVC metric and flow-field tortuosity provide independent support for this interpretation based on coupling.

These conclusions are subject to the limitations of two-dimensional, isothermal Stokes-flow simulations with first-order dissolution kinetics, which do not include precipitation, secondary minerals, multiphase effects, or mechanical deformation. Future work should extend the framework to three-dimensional pore structures, more complex reaction kinetics, and experimental or field NMR calibration. Such validation would help assess whether transient T_2 signatures can monitor evolving hydraulic connectivity during natural and engineered water-rock reactions.

Conflict of Interest

The authors declare there are no conflicts of interest for this manuscript.

Open Research Section

Data statement

The figure source data and supplementary videos supporting this study are available from Zenodo (B. Wang et al., 2026).

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